Ayush Mishra

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SUMMARY

MS Computer Science graduate with a specialty in Network Engineering/Automation and Backend Development. More than 3 years of industry experience in scalable APIs and full-stack development.

EDUCATION

University of Colorado-Boulder **May 2026**
MS, Computer Science (Network Engineering and Backend Development) *Boulder, CO*
Relevant Course Work: Enterprise Networks, Network Systems, Network Automation, IP Routing and Protocols, Network Analysis and modelling

CAPABILITIES

- Comfortable with IP addressing design and implementation.
- Familiar with use of packet analysis tools (Wireshark, TCPdump) to perform protocol analysis and troubleshooting.
- Able to install, configure, and monitor Ethernet technologies and routing protocols for enterprise solutions.
- Hands on experience implementing security protocols like IPSec based VPN services.
- Comfortable using network tools like Netmiko and NAPALM to automate network provisioning and monitoring.
- Experienced in database design and writing software tools to interact with them.
- Proficient with full stack UI application development and versioning tools like Git and BitBucket.
- Able to program scalable APIs for systems processing large amounts of transactions.
- Also, familiarized with latest software development trends, integrating LLMs/AI to all previous domains.

NETWORK EXPERIENCE

Network Deployment

- At least 300 hours of hands-on experience implementing the following:
  - MPLS with OSPF/LDP and MPLS-TE (RSVP-TE/CSPF) for primary/backup path engineering, and Policy-Based Routing.
  - Campus-style Layer 2/3 switching with VLANs, trunking, inter-VLAN routing, and enhanced resiliency using STP/RSTP and EtherChannel; Also IP resiliency using HSRP and ECMP.
  - Deployment of various routing protocols (OSPF, RIP, EIGRP, BGP) for Enterprise and Internet uplink connectivity; and relevant features needed for private to public network communication.
  - Developed software applications for network services in C, including a UDP-based FTP client/server, a concurrent TCP HTTP web server, an HTTP caching and reverse proxy, and a distributed file system that shards files across multiple servers for redundancy.

Network Automation

- Well versed in the use of network automation tools including;
  - Automated multi-router configuration workflows using Python (Netmiko/NAPALM) with external config files, IPv4 validation + reachability checks, and saved device running-config snapshots for repeatable deployments.
  - Deployed network automation tools to collect routing protocol inputs, configuration snapshot storage in database, and generate config diffs to detect drift between current and previously saved configurations.

- Integrate LLMs/AI in automation applications for various network configuration tasks.

SOFTWARE DEVELOPMENT EXPERIENCE

Cisco Systems (Capstone, Work in Progress) **Sep 2025 - Pres.**
Software(Network) Engineer *Boulder, CO*

- Development of an Agentic LLM-driven, intent-based network configuration system, using Python/Flask backend, MongoDB vector search, and React UI.
- Infused Prompt-Engineering, LangChain, Token limitation, and custom tool calling into the application to perform network automation tasks using user prompts.

VTM Group **May 2023–Aug 2024**
Technical Marketing Engineer *Beaverton, OR*

- Worked on centralized platforms to collect and manage test results from various hardware vendors.
- Contributed with development of tools used for testing for PCI-SIG.

Novus Labs **Oct 2021–Apr 2023**
Software Developer *Hillsboro, OR*

- Worked to build solutions for Fortune 500 clients using C++ and Python.
- Built backend for Web/IoT Applications for real-time data storage, and tools for Law-firm using Pacer-API.
- Also, implemented source-to-source code transformation using clang-LibTooling(C++) for ROS

Tokn Inc. (Startup) **May 2021–Jan 2022**
Software Developer (Backend) *Remote*

- Designed backend to use blockchain technologies for the commercialization of music, using NoSQL and Solidity.

TECHNOLOGIES & TOOLS

- **Programming Languages:** C++; Python; Cisco IOS; JS (ReactJs, VueJs); MongoDB; SQL; AWS; Docker; Linux; Laravel;
- **Network Protocols:** IPv4/IPv6; TCP/IP; OSPF; BGP; ARP; DNS; STP; RIP; EIGRP; Wi-Fi; MPLS; IPSec;
- **Network Automation:** Netmiko; NAPALM; Wireshark; GNS3; Jinja2; Ansible; Container Lab; IaC
- **AI Software:** LangChain; RAG; Prompt Engineering; Fine-Tuning; MCP;
- **Version Tools:** Git; BitBucket; Jira;

Certifications

- **Cisco Certified Network Associated** (CCNA**,** Exp. May 2026)
- **JNCIS DevOps** (Exp. June 2026)